Source Code Management using GitHub SOP

Cognizant

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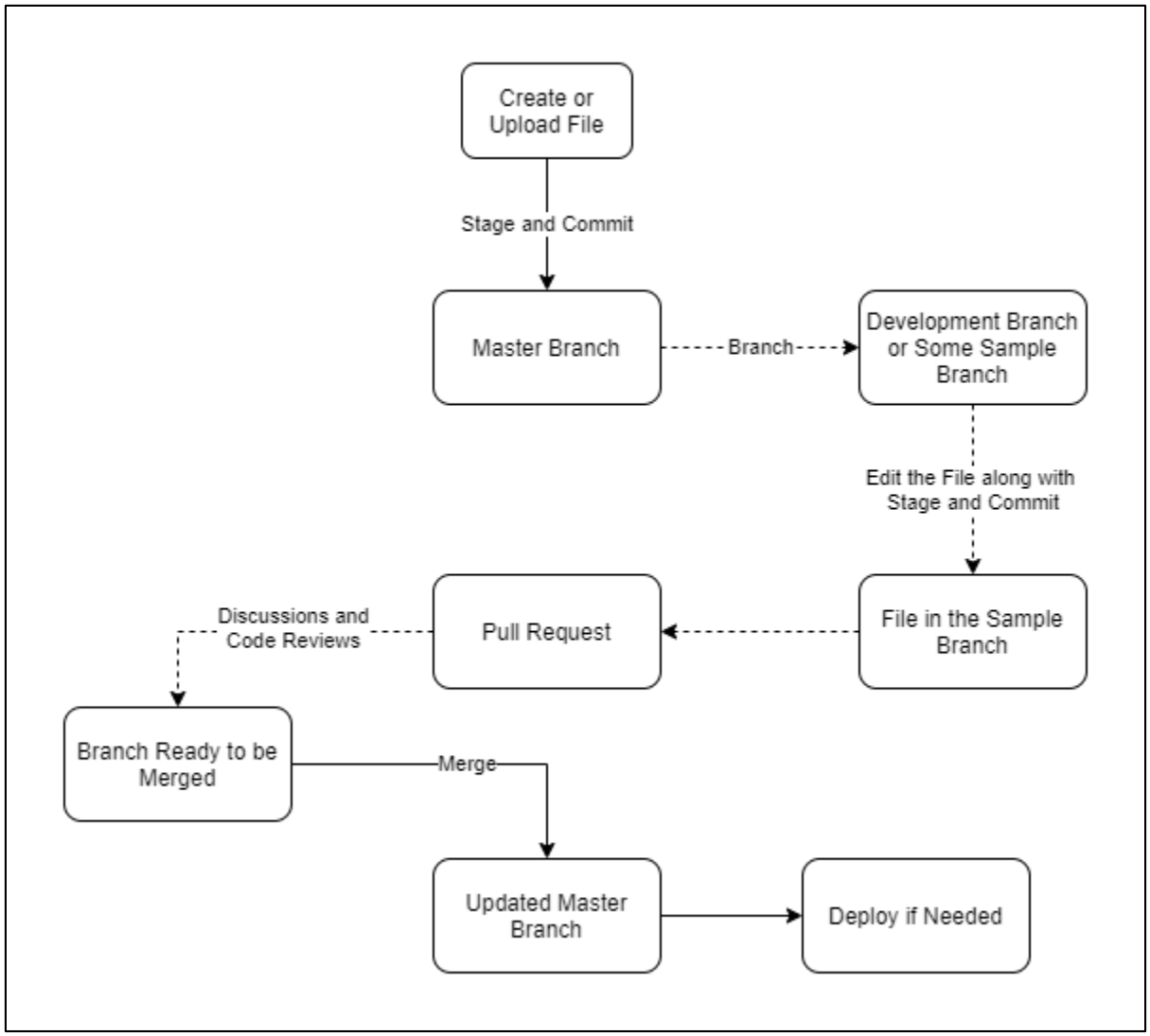
1. Source Code Management using GitHub

1.1 Description

GitHub is a Distributed Version Control System that helps keep track of code changes and maintains repositories. Once the flow of GitHub is understood, it is very easy to work and collaborate on GitHub. In this usecase, a basic workflow is discussed.

1.2 Architecture Diagram

The diagram below displays a visual representation of the underlying application architecture:



1.3 Lab Steps

Follow the steps outlined below to achieve the objective of this lab exercise:

1. Sign up for GitHub using this link. If you already have an account, log in to that.
2. Create a new repository by clicking on **New** in GitHub Home Page.
3. Create a sample repository directly and give some name to it.
   Note: There is no need to add a **README file** or **gitignore file** or **choose a license**.
4. Once the repository is created, you can upload a file by clicking on **Add file** > **upload files**. Or you can create a file then and there using the option **Add file** > **Create new file**.
5. If you selected **upload files** option, drag and drop the file you wish to upload.
6. Once you dropped the files, in the **Commit Changes** section, give some title and description for the upload.
7. Click on **Commit Changes**.
   Note: This process is called staging and committing. If Git Bash is being used, you need to enter the command for stage and commit. In this case it is not required.
   Once Committed, you can see the file in the repository. Now assume this is your master code. Master code should never be touched directly. It should be changed only via **Branches and Pull Requests (PR)**.
8. In our case, create a Branch named **Dev** or something. To perform this, click on **master** and type the name of the branch.
9. Make sure that you are in sample branch. Now edit the file that was uploaded and commit the changes.
10. Once you changed the file, move to **Repository** Home page. Make sure that you are in **Sample Branch**.
11. You should be seeing an option stating, **<branch> had recent pushes Compare and Pull Request**.
12. Click on **Compare and Pull request**. You should be finding **Able to Merge**.
13. Give some title and some description for the PR and create a PR.
14. Now it displays **This branch has no conflicts with the base branch**. Merge Pull request.
15. Click on **merge pull request** and the change will be added to the master branch and click on **delete branch** to delete the branch that was created recently.
16. If you navigate to the repository home, it should display master as the branch. Check the file that was edited and you should find the changes integrated.
    This is the general flow of GitHub Source Control.